Optimal enfranchisement

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Abstract

A planner picks the franchise — the subset of agents allowed to vote — for a binary decision that affects different agents with different intensities. This note revisits the optimal enfranchisement model of Fleurbaey and Lojkine [1] without their symmetry assumption on opinions. Two natural conclusions from the symmetric model fail in full generality, even when the joint opinion distribution has full support: a planner's dictatorship can beat every majority franchise, and the unique optimal franchise need not be composed of the highest-stake agents. We give two five-agent counterexamples.

1 Introduction

Many collective decisions are binary — admit a country, lift a curfew, ban a substance — and bear unequally on the population. A normative theory that takes preference intensity seriously must decide who should vote. We give a stylized model of this trade-off and ask two natural questions:

1. Is voting always at least as good as letting the planner pick an outcome ex ante?
2. Is the optimal franchise the one composed of the highest-stake agents?

Fleurbaey and Lojkine [1] prove that, under ex ante permutation invariance of opinions, universal franchise beats every planner's dictatorship and every optimal franchise is high-stake. They also analyze the common-iid case and the fair-coin prefix problem. This note shows why their symmetry assumption is doing real work. Under full support of the joint opinion distribution alone, the two positive conclusions need not survive: Section 3 gives a five-agent example where every odd majority franchise is beaten by the constant rule $Y \equiv 1$, and a second five-agent example where the unique optimal franchise excludes a strictly higher-stake agent in favour of a lower-stake one.

Related literature. This note is a companion to Fleurbaey and Lojkine [1]. The closest classical results are the Condorcet jury theorems [2], which give independence and competence conditions under which majority correctness rises with the size of a homogeneous jury. The expert-voting literature [3, 4] solves for optimal weighted majority rules when voter competences are observed, with weights $\log(p_i/(1-p_i))$. Schmitz and Tröger [5] study ex-ante utilitarian welfare for binary alternatives with uncertain preference intensities; their main message agrees with the one here — symmetry and neutrality make majority attractive, independence restores strong optimality, and correlated utilities can make majority suboptimal. The Penrose–Banzhaf voting-power tradition [6, 7] stresses that a voter's marginal influence is governed by the probability of a tie among the others. For schemes that aggregate intensity directly rather than choosing a franchise, see Casella and Gelman [8].

2 Model

A planner faces a population $\mathcal{N} = \{1, \ldots, N\}$ and a binary decision $\{0, 1\}$. From her vantage, agent i's opinion is a Bernoulli random variable X_i. The planner picks a *franchise* $C \subseteq \mathcal{N}$ and the outcome is the strict majority within C:

$$Y_C = \mathbf{1}_{\{S_C > |C|/2\}}, \qquad S_C = \sum_{i \in C} X_i.$$

Strict majority is unbiased only when $|C|$ is odd, so we assume throughout that N is odd and consider only nonempty franchises of odd size. Let $p(i, C) = P[X_i = Y_C]$.

Each agent has utility $U(i, C)$ that depends on her own success.

Assumption 1 (Selfishness)**.** Conditional on $\mathbf{1}_{\{X_i = Y_C\}}$, agent i is indifferent to the success of others and to anyone's inclusion in the franchise.

Under Assumption 1, $\mathbb{E}[U(i, C) \mid X_i = Y_C]$ and $\mathbb{E}[U(i, C) \mid X_i \neq Y_C]$ are constants in C. Define

$$\Delta_i = \mathbb{E}[U(i, C) \mid X_i = Y_C] - \mathbb{E}[U(i, C) \mid X_i \neq Y_C] \geq 0,$$

the agent's *stake*. Label agents so $\Delta_1 \geq \Delta_2 \geq \cdots \geq \Delta_N \geq 0$ and drop the constant terms from utility. Aggregate expected welfare under franchise C is

$$U(C) = \sum_{i \in \mathcal{N}} \Delta_i \, p(i, C). \tag{1}$$

Assumption 2 (Full support)**.** For every $x \in \{0, 1\}^N$, $P[(X_1, \ldots, X_N) = x] > 0$.

For comparison we consider a *planner's dictatorship*: an outcome $Z_q \sim \text{Bernoulli}(q)$ independent of $(X_1, \ldots, X_N)$, giving welfare $V(q) = \sum_{i \in \mathcal{N}} \Delta_i \, P[X_i = Z_q]$. The objective is linear in q, so the best dictatorship is either $q = 0$ or $q = 1$.

Definition 1 (High-stake franchise)**.** $C \subseteq \mathcal{N}$ is *high-stake* if $C = \mathcal{N}$ or $\Delta_i \geq \Delta_j$ for every $i \in C$ and every $j \in \mathcal{N} \setminus C$.

3 Two negative results

3.1 Dictatorship can beat every franchise

Proposition 1. *There exist stakes $(\Delta_1, \ldots, \Delta_5)$ and a full-support distribution under which every odd franchise has strictly lower expected welfare than the dictatorship $q = 1$.*

Proof. Take $N = 5$ and $(\Delta_1, \ldots, \Delta_5) = (5, 4, 3, 2, 1)$. Define P_0 on $\{0, 1\}^5$ by the following four atoms, with integer weights summing to 6:

x	$6\,P_0[X=x]$
$(0,1,1,1,0)$	1
$(1,0,0,1,1)$	3
$(1,0,1,0,1)$	1
$(1,1,0,0,0)$	1.

The dictatorship $q=1$ has welfare $17/2$. The welfares of all sixteen odd franchises, multiplied by 6, are:

C	$6\,U_{P_0}(C)$	C	$6\,U_{P_0}(C)$
$\{1\}$	48	$\{1,2,3\}$	48
$\{2\}$	45	$\{1,2,4\}$	48
$\{3\}$	45	$\{1,2,5\}$	48
$\{4\}$	45	$\{1,3,4\}$	48
$\{5\}$	45	$\{1,3,5\}$	45
$\{1,4,5\}$	45	$\{2,3,4\}$	42
$\{2,3,5\}$	45	$\{2,4,5\}$	45
$\{3,4,5\}$	48	$\{1,2,3,4,5\}$	48.

The best franchise welfare is 8, so the dictatorship beats every franchise by at least $1/2$.

Mix P_0 with the uniform distribution R on $\{0,1\}^5$ to enforce full support:

$$P = \tfrac{99}{100}\,P_0 + \tfrac{1}{100}\,R.$$

Total stake is 15, so replacing P_0 by P can shrink any welfare gap by at most $15/100$. Since

$$\tfrac{99}{100}\cdot\tfrac{1}{2} \;-\; \tfrac{15}{100} \;=\; \tfrac{69}{200} \;>\; 0,$$

the dictatorship $q=1$ still beats every franchise under the full-support distribution P. $\square$

3.2 Optimal franchises need not be high-stake

Proposition 2. *There exist strictly decreasing stakes and a full-support distribution under which the unique optimal franchise is not high-stake.*

Proof. Take $N=5$ and $(\Delta_1,\ldots,\Delta_5)=(5,4,3,2,1)$. Define P_0 on three states:

x	$P_0[X=x]$
$(1,0,0,0,1)$	$1/5$
$(1,0,0,1,1)$	$3/5$
$(1,1,0,0,0)$	$1/5.$

Let $C^\star = \{1,2,4\}$. The welfares $5\,U_{P_0}(C)$ across the sixteen odd franchises are:

C	$5\,U_{P_0}(C)$	C	$5\,U_{P_0}(C)$
$\{1\}$	39	$\{1,2,3\}$	39
$\{2\}$	39	$\{1,2,4\}$	42
$\{3\}$	36	$\{1,2,5\}$	39
$\{4\}$	39	$\{1,3,4\}$	39
$\{5\}$	36	$\{1,3,5\}$	36
$\{1,4,5\}$	36	$\{2,3,4\}$	36
$\{2,3,5\}$	36	$\{2,4,5\}$	39
$\{3,4,5\}$	39	$\{1,2,3,4,5\}$	39.

$C^\star$ is the unique maximizer, beating every other franchise by at least 3/5.

Mixing with the uniform R on $\{0,1\}^5$,

$$P = \tfrac{99}{100}\,P_0 + \tfrac{1}{100}\,R,$$

gives full support. Total stake is 15, so for any other franchise D,

$$U_P(C^\star) - U_P(D) \;\geq\; \tfrac{99}{100}\cdot\tfrac{3}{5} \;-\; \tfrac{1}{100}\cdot 15 \;=\; \tfrac{111}{250} \;>\; 0,$$

and $C^\star$ remains the unique optimum. Because the stakes are strictly decreasing, the only high-stake franchises of odd size are $\{1\}$, $\{1,2,3\}$, and $\{1,2,3,4,5\}$. Since $C^\star$ excludes agent 3 (stake 3) while including agent 4 (stake 2), $C^\star$ is not high-stake. □

Intuition. In Proposition 2, agent 3's opinion is constantly 0 under P_0 and so conveys no information, while agent 4's opinion marks the high-probability state in which outcome 1 best serves the high-stake agents. Swapping voter 3 for voter 4 in the size-three franchise changes Y in exactly the state $(1,0,0,1,1)$, flipping it from 0 (pleasing stakes $4+3=7$) to 1 (pleasing stakes $5+2+1=8$); the net gain $1\times 3/5$ is the entire margin. In Proposition 1, the joint distribution is biased towards $X_1 = 1$, and the constant dictator who sides with the highest-stake agent outperforms any majority of a fluctuating subset.

4 Relation to existing positive results

The two counterexamples above are outside the symmetric-opinions domain analyzed by Fleurbaey and Lojkine [1]. In that paper, full support plus ex ante permutation invariance of $(X_i)_{i\in\mathcal{N}}$ is enough to prove that universal franchise has strictly larger expected aggregate welfare than any planner's dictatorship, and that every optimal franchise is high-stake. Their notation writes the success probabilities as $p_+(c)$ for insiders and $p_-(c)$ for outsiders in a franchise of size c; their key lemma is $p_+(c) > p_-(c)$, which immediately yields high-stake optimality by exchanging a lower-stake insider for a higher-stake outsider.

The iid cases that one might use as positive benchmarks here are therefore already covered there. When opinions are common iid Bernoulli, Fleurbaey and Lojkine derive the fixed-size comparison criterion and the associated one-dimensional search over high-stake prefixes; in the balanced case $p = 1/2$, this gives the familiar central-binomial term governing whether a larger prefix is worthwhile. This note does not reprove those results. Its purpose is instead to show that, once permutation invariance is dropped, full support alone is too weak: the dictatorship comparison and the high-stake conclusion can both fail.

References

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